# Technology and AI Use Declaration

SDC Hackathon 2026 | Team 牛马来西亚

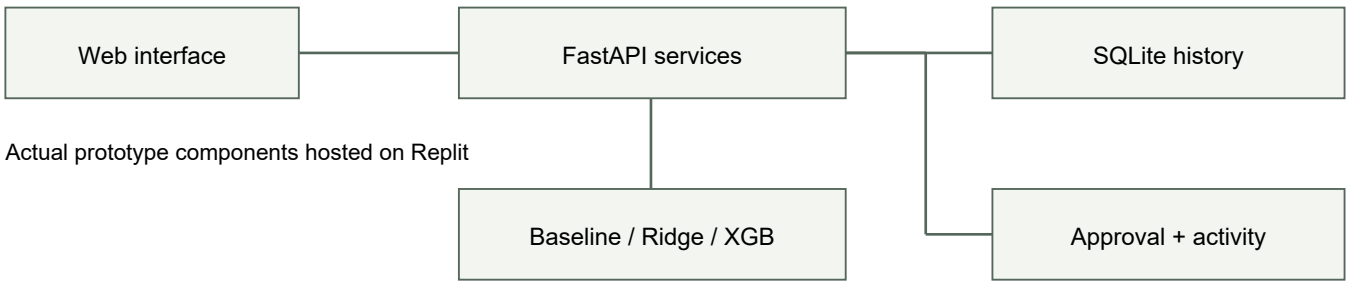
**Prepared against all sections of the official SDC technology declaration template. Factual disclosures are completed from project records; the leader must review the checklist and sign before formal submission.**

## 1 Team and Project Information

| Field | Response |
|---|---|
| Team / project | 牛马来西亚 / PrepWise |
| Domain / SDG | Domain 2 - Orbyte Smart SME Challenge / SDG 12.3; supporting productivity objectives under SDG 8 |
| Leader | Lan Dongao |
| Leader contact | lansoar000@gmail.com; +60 1123318006 |
| Institution / team size | Xiamen University Malaysia / 5 |
| Members | Shi Jiahuan; Cui Liuyang; Chen Yan; Wei Shangao |
| Live prototype | https://sdc-hackathon-2026--lansoar000.replit.app/ |
| Public repository | https://github.com/lansoar000-code/SDC-Hackathon-2026 |

## 2.1 System Architecture Overview

The web interface sends requests to FastAPI on Replit. The backend validates imports, stores dish sales and forecast snapshots in SQLite, and calls the forecasting module. Historical baseline, Ridge and XGBoost are compared using time-based validation when sufficient history is available. Model artifacts are stored as JSON. A user-selected buffer produces preparation suggestions. The operator can approve a separate final plan, which records an approval timestamp and application events. The current interface uses simulated identity and is suitable only for synthetic demonstrations. The public demo at the evidence check used fixed rules; the live dataset was empty and no trained model was loaded. A future Orbyte connector is proposed but not implemented.



## 2.2 Languages Frameworks and Libraries

| Name / version | License / purpose |
|---|---|
| Python 3.12 | Python Software Foundation license; backend runtime |
| FastAPI 0.159.1 / Pydantic 2.14.1 | MIT; APIs and input validation |
| Uvicorn 0.54.0 / Starlette 1.9.0 | BSD-3-Clause; HTTP serving and web framework |
| openpyxl 3.1.5 | MIT; XLSX import |
| python-multipart 0.0.32 | Apache-2.0; multipart file uploads |
| NumPy 2.5.4 | BSD-3-Clause, 0BSD, MIT, Zlib and CC0-1.0 bundled notices; numerical methods |
| XGBoost 3.4.1 | Apache-2.0; gradient boosted tree training |
| SQLite | Public domain; local application database |

Direct package versions agree with the repository requirements; transitive versions above are from the checked local environment. Exact deployed transitive versions were not independently captured. The complete lockfile also contains transitive and testing dependencies.

## 3.1 External APIs

No external generative-AI, weather, payment, messaging or remote inference API is called by the audited forecasting implementation. Internal /api routes are the team's own backend APIs. Google Fonts and an external food photograph are browser resources, not forecasting inputs.

## 3.2 Cloud Platforms and Deployment

| Service / provider | Purpose and status |
|---|---|
| Replit | Public frontend and FastAPI hosting. Publishing status success was checked on 25 October 2026. The publishing interface showed Starter/free deployment, Autoscale and North America. Free deployment expiry applies; durable storage across republishing is unverified. |
| GitHub | Source repository and submission documents. Repository is public; source code and submission documents are accessible. |
| Google Drive | Final PPTX and complete 6:05.4 demonstration video uploaded. Both allow anyone with the link to view. Account is a personal Google Drive account; storage region was not recorded. |
| Cloudflare Quick Tunnel | Earlier temporary local preview only; not the production deployment. |
| RunPod | Not used, as confirmed by the team. No cloud workload or sponsor API integration is claimed. |

## 4.1 AI Models Used in the Solution

| Model | Source / usage / license |
|---|---|
| Historical baseline | Team implementation using past weekday sales and a recent-history fallback; project code Apache-2.0 |
| Ridge regression | Team implementation using NumPy; trained from synthetic records when historical mode is used; project code Apache-2.0 plus NumPy notices |
| XGBoost | dmlc/XGBoost framework; self-trained on team synthetic records; Apache-2.0 |
| Fixed-rule demo | Explicit synthetic illustration; not a trained AI model and excluded from performance metrics |

## 4.2 Datasets

Team-generated synthetic cafeteria/restaurant dish histories, with reproducible generation settings. The larger local dataset contains 80,000 dish-date rows across 10,000 dates and eight fictional dishes. These dates are fictional data labels, not a claim that real records were collected over that period. Personal/customer/company records are not included in the declared dataset. Third-party dataset: none declared. The generator is team project code under Apache-2.0. No separate third-party dataset license is applicable; no separate license for generated outputs has been assigned.

## 4.3 AI Assisted Development Tools

| Tool / extent | Use and disclosure |
|---|---|
| ChatGPT / Codex - major | Code, debugging, documentation, UI and test assistance. Exact model/version per contribution was not completely recorded. |
| Replit Agent - moderate | Project setup, coding and deployment troubleshooting assistance. Some requested tasks did not complete because of quota exhaustion. |
| OpenAI image generation - minor | Earlier interface concept/reference artwork. It informed design; the deployed hero is an external Unsplash photograph. Exact model version and asset-by-asset use in the supplied final deck/video were not recorded. |

Review process: AI-assisted changes were checked through automated regression tests and observed public-demo/API checks. The approval integration passed 33 regression tests, including eight approval cases. The public workflow generated and approved a labelled synthetic plan. These checks establish the tested behavior, not complete correctness or measured business impact. The leader must review this declaration and confirm responsibility for the submitted work before signing; individual member review coverage was not recorded.

## 5 Official Sponsor Resources

| Sponsor / resource | Actual use |
|---|---|
| Replit / cloud platform | Hosts the functional prototype. Real public screenshots and saved synthetic approval evidence are in the Replit proof PDF. |
| RunPod / voucher | Voucher availability is not usage. Team confirms no actual use; do not claim completed integration or proof. |
| API Premium eligibility | Unresolved. The portal requires a sponsor-integration declaration. No assertion of eligibility or waiver is made here. |

Replit is the current hosting platform. The core CPU forecasting code could run on another Python-compatible cloud host. Replit supplies the present deployment and sharing workflow, rather than a unique forecasting algorithm.

### 6.1 Data Handling and Privacy

The app can store uploaded dish sales, forecasts, final plans and activity events in SQLite. The declared demonstration uses synthetic records. The interface currently has no secure login or separation between businesses, so real customer or confidential business records should not be used. No such records are sent to an external LLM by the audited implementation. Automated retention/deletion controls and cloud durability have not been established. A real deployment would need access control, data minimisation, retention rules and tenant separation before processing real records.

### 6.2 Known Limitations and Risks

Synthetic evaluation does not establish real restaurant accuracy or waste reduction. The inspected synthetic evaluation period has been reused and should not be described as a new independent test. Fixed-rule demo output is not a trained-model result. At the evidence check, the live database had zero sales dates and its model was untrained. Cloud persistence across republishing remains unverified. The owner identity is simulated, not authenticated. Recipe/inventory purchasing automation and a live Orbyte connector are not implemented in the deployed prototype. Team targets for operational savings require a defined baseline and comparable evaluation.

## 7 Compliance Checklist and Leader Declaration

| Official checklist item | Review status |
|---|---|
| All resources fully declared | Main resources disclosed; final PPT/video asset provenance is not fully recorded and requires leader review |
| Licenses and attribution satisfied | Team review required; preserve third-party notices and verify external image/font sources |
| No exposed credentials | Current 80 text files scanned for common secret patterns; no matches or tracked databases found. This is not a full historical credential audit. |
| AI assistance disclosed and reviewed | Tools disclosed; leader responsibility confirmation and signature required |
| Sponsor resources declared | Replit use disclosed; RunPod non-use disclosed; qualifying sponsor requirement unresolved |
| Judges can access prototype and repository | Public prototype, repository, PPT and video links are provided; accessibility checks were performed |
| All fields and word limits complete | All template categories addressed; leader checklist confirmation and signature remain |

## Additional asset disclosure

DM Sans and Libre Caslon Display are loaded through Google Fonts. An Unsplash food photograph is used in the interface. Verify the exact font license notices, photograph source/author and the sources of final PPT/video images before certifying the checklist. Qualitative METMalaysia climate information informed the synthetic-data context; it is not a live weather API.

## Leader checklist - complete before signing

Leave a box empty unless you can confirm the statement. This prepared copy does not certify unchecked statements.

[ ] All APIs, cloud services, AI tools, models and datasets used are declared.
[ ] Third-party licenses and required attribution have been satisfied.
[ ] No credentials are exposed in the repository, prototype or this document.
[ ] Generative-AI assistance has been disclosed and reviewed by the team.
[ ] Official sponsor resources are declared in Section 5.
[ ] The prototype and repository are accessible to judges.
[ ] All fields are complete and answers meet the specified word limits.

A signature in the separate proposal does not sign this declaration.

## Submission material links

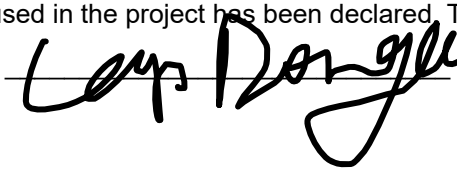
PPT: https://docs.google.com/presentation/d/1vcyDAXWv9w9mDHFz-JKE3pWYl4U1rePp/edit

Video: https://drive.google.com/file/d/1JNB7649TAayQP9EpgzflGmFhdiUSBBDp/view

UI photograph source: https://images.unsplash.com/photo-1512058564366-18510be2db19. Font distribution: Google Fonts. Additional development/document tools: Ruff, unittest, HTTPX, ReportLab, pypdf and PDFium. FFmpeg (via imageio-ffmpeg 0.6.0) produced the complete smaller video upload copy; the original video was retained. These are tooling resources, not runtime AI services.

## Leader declaration and signature

Leader: Lan Dongao. By signing, the leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model and sponsor resource used in the project has been declared. The signature field is reserved for the leader.

Signature:  Date: 